COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	July 17-21, 2026	**DATE SUBMITTED:**	July 21, 2026
NAME:	Jaime Luis M. Demain	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

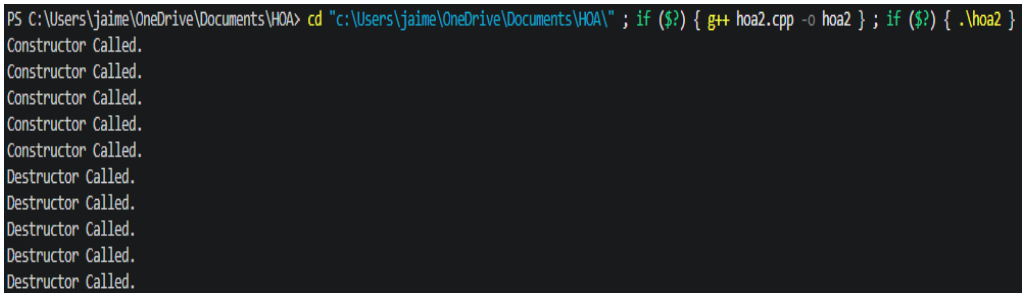
ACTIVITY NO. 2.1 **Arrays, Pointers and Dynamic Memory Allocation**

1. PROCEDURE OUTPUT

What procedures did you perform?

Screenshot	
Observation	The output have effectively illustrated how the lifecycle of an object works and how 'Big Three' member functions get implemented. The first constructor has initialized student1, whereas the first copy constructor has created student2 from it. student3 gets created using the second constructor. Whenever student3 = student2; the statement executes, and our copy constructor will execute once again because of the assignment operator since it takes the object parameter by value and is followed by our copy assignment function. Four destructor statements have been executed: one for the temporary parameter object of the assignment operator, and three more for our local objects.

Table 2-1

Screenshot	

Observation	In this part, we see how the initialization of the static array invokes the default constructor for each individual item in the array, based on the size of the array. Initialization of the array named studentList with a size of 5 leads to automatic invocation of the default constructor five times to initialize the array items on the stack. These helper arrays namely namesList and ageList have been initialized and are not dependent on the student array yet. Lastly, on exiting the main() scope, 5 destructors will be called automatically to de-allocate the stack memory used by the array.

Table 2-2

Loop A	`for(int i = 0; i < j; i++){ //loop A`
	`Student *ptr = new Student(namesList[i], ageList[i]);`
	`studentList[i] = *ptr;`
	`}`

Observation	In Loop A, there is an example of dynamic memory allocation with deep copying into a static array. Each time when a new Student object is created using the new operator, the corresponding constructor is activated. On assignment to a static array, the copy constructor is automatically invoked because of passing a value, followed by copying and destroying the temporary parameter object.

Loop B	`for(int i = 0; i < j; i++){ //loop B`
	`studentList[i].printDetails();`
	`}`

Observation	Loop B prints the member data of each object in the array, using printDetails(), which shows that the data from the heap objects have been successfully copied to the static array stack elements through Loop A.

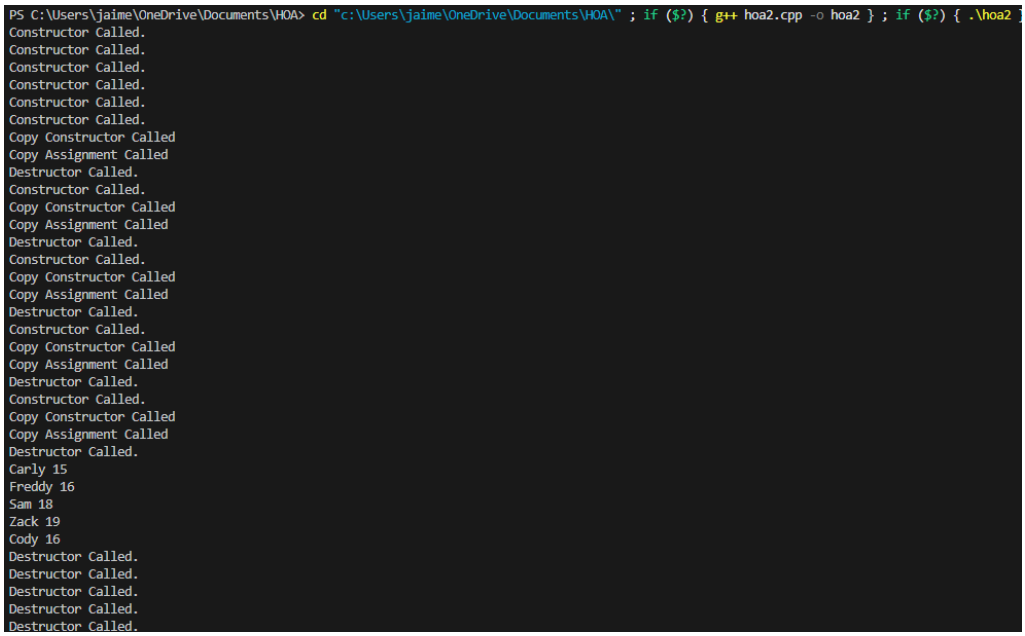
Output	
Observation	Output produced in the end has been correctly listed with updated profiles of students. But the log files show a clear case of memory leak since there is an observed difference between the number of constructors invoked that is 10 (5 static and 5 dynamic) and the number of final destructors used for the stack objects that is 5.

Table 2-3

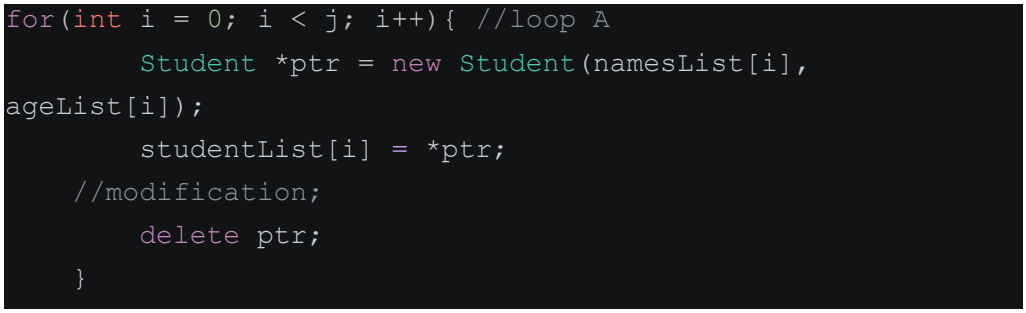
Modifications	
Observation	The memory leak detected at Loop A was fixed by inserting delete ptr. Since the original dynamically allocated object Student (ptr) has served its purpose of transferring the initialized properties to the static studentList array, not releasing it led to orphan memory on the heap. Using delete operator guarantees that the destructor will be invoked as soon as the value has been transferred for every dynamically allocated new. This guarantees that the heap memory is managed efficiently and leads to a balanced number of constructors and destructors, 10 each at runtime exit.

Table 2-4

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve programming problems using dynamic memory allocation, arrays and pointers
Jenna's Grocery List
Apple PHP 10 x7
Banana PHP 10 x8
Broccoli PHP 60 x12
Lettuce PHP 50 x10
Jenna wants to buy the following fruits and vegetables for her daily consumption. However, she needs to distinguish between fruit and vegetable, as well as calculate the sum of prices that she has to pay in total.
Problem 1: Create a class for the fruit and the vegetable classes. Each class must have a constructor, deconstructor, copy constructor and copy assignment operator. They must also have all relevant attributes (such as name, price and quantity) and functions (such as calculate sum) as presented in the problem description above.
Problem 2: Create an array GroceryList in the driver code that will contain all items in Jenna's Grocery List. You must then access each saved instance and display all details about the items.
Problem 3: Create a function TotalSum that will calculate the sum of all objects listed in Jenna's Grocery List.
Problem 4: Delete the Lettuce from Jenna's GroceryList list and de-allocate the memory assigned.
Problem 5: Discuss the

```
PS C:\Users\jaime\OneDrive\Documents\HOA> cd "c:\Users\jaime\OneDrive\Documents\HOA\" ; if ($?) { g++ hoa2supp.cpp -o hoa2supp } ; if ($?) { .\hoa2supp }
Constructor Called.
Constructor Called.
Constructor Called.
Constructor Called.

--- Populating Fruit List via Dynamic Memory ---
Constructor Called.
Copy Assignment Called
Destructor Called.
Constructor Called.
Copy Assignment Called
Destructor Called.

--- Populating Vegetable List via Dynamic Memory ---
Constructor Called.
Copy Assignment Called
Destructor Called.
Constructor Called.
Copy Assignment Called
Destructor Called.

--- Displaying Jenna's Grocery List ---
[Fruit] Apple | PHP 10 | x7 | Subtotal: PHP 70
[Fruit] Banana | PHP 10 | x8 | Subtotal: PHP 80
[Vegetable] Broccoli | PHP 60 | x12 | Subtotal: PHP 720
[Vegetable] Lettuce | PHP 50 | x10 | Subtotal: PHP 500

Initial Grand Total: PHP 1370

--- Removing Lettuce from List ---
Constructor Called.
Copy Assignment Called
Destructor Called.

New Grand Total after removing Lettuce: PHP 870

--- End of Main Scope (Stack Cleanup) ---
Destructor Called.
Destructor Called.
Destructor Called.
Destructor Called.
PS C:\Users\jaime\OneDrive\Documents\HOA>
```

Output of the Supplementary Activity

Conclusion: This activity helped me understand practical concepts regarding lifetimes of objects, dynamic memory allocations in the heap, and the importance of "Big Three" in C++. By following the steps listed in the procedure, I learned how the initializations of arrays statically would lead to the invocation of default constructors, and that through the use of deep copying, one can easily experience memory leaks in the heap without properly releasing memory in the heap through the use of "delete". This was enhanced through the application of the above knowledge in the supplementary activity part where I used separate object classes and dynamic pointers to effectively handle memory leaks while making calculations on total totals.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance for copy assignment operator part in procedure and supplementary activitiy. Also, it helped me to detect any memory leaks on the final code of procedure and finalize my whole output in supplementary activity. Then, improvement of grammar in observation and conclusion.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used gemini to visualize how copy assignment can be made so that I gain ideas to form it. Also, I used AI to help me understand the supplementary activity; furthermore, it helped me accomplish it as it is kinda complicated for me to finish the whole output. And to detect any possible improvements of the final code in Procedure and Supplementary part. Plus, I used the AI to improve my grammar in forming an observation and conclusion and correct its grammar mistakes.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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